AWS Fault-Tolerant Portfolio Hosting Project Guide

Architecture Diagram

```
     +-------------------+
     |       Users       |
     +---------+---------+
               |
               v
     +-------------------+
     |    CloudFront     |
     |       (CDN)       |
     +---------+---------+
               |
               v
     +-------------------+
     | Application Load  |
     |     Balancer      |
     +---+-----------+---+
         |           |
         v           v
+-------------+  +-------------+
| EC2 AZ-1    |  | EC2 AZ-2    |
| Nginx       |  | Nginx       |
+-------------+  +-------------+

     CloudWatch Monitoring
             |
             v
  EC2 + ALB Metrics/Alarms
```

Project Objective

Host a portfolio website on two EC2 instances across different Availability Zones, place them behind an Application Load Balancer, accelerate delivery using CloudFront CDN, and demonstrate failover when one server becomes unavailable.

Phase 1 – Create Networking

Create VPC, Public Subnets in AZ1 and AZ2, Internet Gateway, Route Table, and associate both public subnets.

Phase 2 – Security Groups

ALB SG: HTTP 80 from Anywhere. EC2 SG: SSH 22 from My IP and HTTP 80 from ALB SG.

Phase 3 – Launch EC2 Instances

Launch two Ubuntu t3.micro instances in different Availability Zones.

Phase 4 – Install Nginx

Install nginx, start the service, and enable it on boot.

Phase 5 – Deploy Portfolio Website

Copy portfolio HTML and CSS files into /var/www/html and verify using the EC2 public IP.

Phase 6 – Create Target Group

Create an HTTP target group on port 80 with health check path '/'. Register both instances and wait for healthy status.

Phase 7 – Create Application Load Balancer

Create an Internet-facing ALB, configure HTTP:80 listener, and forward traffic to the target group.

Phase 8 – Configure CloudFront CDN

Create a CloudFront distribution with ALB DNS as origin and wait for deployment.

Phase 9 – Configure CloudWatch

Create CPUUtilization and StatusCheckFailed alarms for monitoring.

Phase 10 – Failover Demonstration

Stop one EC2 instance and verify that the website remains available through the healthy instance.

Troubleshooting

503 = No healthy targets. 504 = Backend not reachable. Unhealthy targets usually indicate nginx, health check, or security group issues.